Formulas and Equations

$c = 310^8\, m/s$ $c = f_0\lambda$ $P_{av} = P_t\,\tau/T_p$ $E_b = P_t\tau$	$[n] = \frac{\theta_B f_p}{\dot{\theta}_s} = \frac{\theta_B f_p}{6\omega_r}$ $G = \frac{26000}{\theta_B \varphi_B[\text{degree}]}$	$R_{max}^4 = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 S_{min}} = \frac{P_t G A_e \sigma}{(4\pi)^2 S_{min}}$ $= \frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 S_{min}}$
$\theta_B = \frac{\lambda\sqrt{\frac{\pi}{4}}}{D}$ or $R = cT_p/2$ $\Delta R = c\,\tau/2$ $\Delta XR_{A/E} = R\ \cdot \theta_{A/E}$ $T_0 = 290$ K	$F_n = \frac{S_{\text{in}}/N_{\text{in}}}{S_{out}/N_{out}}$ $N = kTB$ $d = R \cdot \theta_B$ $\text{GSD} = \text{IFOV} \cdot R$ $P_d = P_t/\text{Area}$	$E_i(n) = (S/N)_n = (S/N)_i/n$ $\lvert H(f)\rvert = 2sin(\pi f_d T_p)$ $\lvert H(f)\rvert = 4sin^2(\pi f_d T_p)$ $\text{IFOV} = \frac{p}{f}$ $A_{ill} = \Omega R^2$
$G = \frac{4\pi A_e}{\lambda^2}$ 1 pico $= 10^{-12}$ 1 nano $= 10^{-9}$ 1 MHz $= 10^6$ Hz 1 MW $= 10^6$ W 1 kW $= 10^3$ W 1 nmi = 1852 m $k = 1.38 \times 10^{-23}$J/K 1knot = 0.5m/s $P_{fa} = \frac{1}{T_{fa}B} = \frac{1}{n_f}$ $P_r = \frac{P_t G_t A_e}{4\pi R^2}$	$T_{fa} = \frac{1}{B} exp\left(\frac{V_T^2}{2\psi_0}\right)$ $A_e = \rho A$ $f_d = 2\frac{v_r}{\lambda}$ $v_r = \frac{n\lambda}{2T_p}\ \ n = 1,2,\ldots$ $\theta_B = \frac{1.22\lambda}{D} rad.$ $\Delta\phi = \frac{2\pi d}{\lambda}\sin(\theta)$ $A_{Rx} = \pi D^2/4$ $SNR_{nonfluct} + L_f(dB) = SNR_{fluct}\ @P_d$	$R_{\max}^4 = \frac{P_t G A_e \sigma}{(4\pi)^2 k T_0 B F_n (S/N)_{\min}}$ $S/N = \frac{P_t G A_e \sigma}{(4\pi)^2 k T_0 B F_n R^4}$ $R_{max}^4 = \frac{P_t G A_e \sigma n E_i(n)}{(4\pi)^2 k T_0 B F_n (S/N)_l}$ $R_{max}^4 = \frac{E_p G A_e \sigma n E_i(n)}{(4\pi)^2 k T_0 F_n (B\tau)(S/N)_l}$ $R_{max}^4 = \frac{P_{av} G A \rho_a \sigma n E_i(n) F^4 e^{-2\alpha R_{max}}}{(4\pi)^2 k T_0 F_n (B\tau) f_p (S/N)_l L_f(n_e) L_s}$ $P_d = \frac{P_t}{\Omega R^2}$ $Area = \pi r^2 = \pi(D/2)^2$